

Fig. 2: An actual-size, single-side PCB for crystal-controlled single-frequency FM transmitter

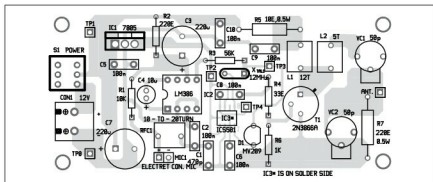


Fig. 3: Component layout for the PCB

signal in your FM receiver at 96MHz.

Construction and testing

An actual-size, single-side PCB for crystal-controlled single-frequency FM transmitter is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable plastic case.

Fix 2-pin connector CON1 for power supply. Use a suitable heat-sink for transistor 2N3866A. Use a $\frac{1}{4}$ -wavelength antenna, which should be around 78cm tall, for 96MHz.

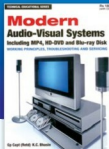
You can construct the circuit even on a piece of copper-clad PCB. The copper cladding acts as the ground plane, giving an excellent performance. Pair the transmitter with a high-sensitivity receiver for a longer range of application.

Before using the circuit, verify the voltages and signal at the test points, as mentioned in the table. ●

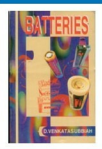
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